DSA ki Baithak

CODE BAITHAK

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Recap (Previous Lecture Connection):

"Pattern Printing Series

with

Interview Questions"

What is Number System ?

It is a way to <u>represent</u> <u>numbers</u> using <u>digits</u> & <u>rules</u>.

<u>Ex</u>: Decimal (Base 10)

Binary (Base 2)

Octal (Base 8)

Hexadecimal (Base 16)

(Can be any base)

What is Base (Radix) ?

It is the total no. of digits used in a Number System.

System	Base	Digits
Decimal	10	0-9 (10)
Binary	2	0, 1 (2)
Octal	8	0-7 (8)
Hexadecimal	16	0-9, A-F (16)

3, 4 $n \rightarrow$ n digits (0, n-1)

10 → A
11 → B
12 → C
13 → D
14 → E
15 → F

Addition :

(Base 10)

$$\begin{array}{r} 1\ \ \ 1\ \ \ \ \\ 1\ 9\ 2\ 8 \\ +\ 3\ 8\ 1\ 5 \\ \hline 5\ 7\ 4\ 3 \end{array}$$

(0-4)

(Base 5)

$$\begin{array}{r} 1\ \ \ 1\ \ \ \ \\ 4\ 4\ 3 \\ +\ 1\ 2\ 3 \\ \hline 1\ 1\ 2\ 1 \end{array}$$

$n \rightarrow 0, n-1$

$\% n \rightarrow (0, n-1)$

$\% 5 \rightarrow 0, 1, 2, 4$

% Base

$$\begin{array}{r} 8\ \ \ \ \

Addition (in Any Base System)

General Addition logic :

* Add digits & carry (sum)
* Result = sum % Base → $n \to (0, n-1)$
* Carry = sum / Base

$8 \% 8 \to 0$
$8 / 8 \to 1$

$11 \% 8 \to 3$
$11 / 8 \to 1$

(0, 7)

(Base 2)
↓
Binary

$$
\begin{array}{r}
1\ 1\ 1\ 1\ 1\ \ \\
1\ 1\ 1\ 1\ 0\ 1 \\
+\ \ 1\ 1\ 0\ 0\ 1\ 1 \\
\hline
1\ 1\ 1\ 0\ 0\ 0\ 0
\end{array}
$$

$2 \% 2 = 0$
$2 / 2 = 1$

(Base 8)

$$
\begin{array}{r}
1\ \ 1\ \ \\
5\ \ 7\ \ 6 \\
+\ \ 5\ \ 3\ \ 2 \\
\hline
1\ \ 3\ \ 3\ \ 0
\end{array}
$$

Decimal Number Expansion

$3 \times 10^1 \rightarrow 30$

$1234 \longrightarrow 1 \times 1000 + 2 \times 100 + 3 \times 10 + 4 \times 1$

(Base 10)

$1 \times 10^3 + 2 \times 10^2 + 3 \times 10^1 + 4 \times 10^0$

1, 2, 3, 4

10^0

(Every digit has a position value)

Positional Value Concept

$$\begin{array}{r} 1000 \\ +\ 200 \\ 30 \\ 4 \end{array} \longrightarrow 1234$$

Conversion Types

* Decimal (Base 10) → Any base System
* Any base System → Decimal (Base 10)
* Any base System → Any base System

(Source base) (Destination base)

Decimal to Any base System :

(0,1)
↑
(Base 2)

Repeaded division by base

$\left[\begin{array}{l} n \% \text{ base} \rightarrow \text{digit} \\ n = n/\text{base} \end{array}\right]$

$(53)_{10} = (\ \ )_2$

$(110101)_2$

①

2	53	1
2	26	0
2	13	1
2	6	0
2	3	1
2	1	1
	0	

10^0
10^1
10^2
10^3
10^4
10^5

1
100
10000
100000
110101

123 → 100 + 20 + 3

$1\times10^2 + 2\times10^1 + 3\times10^0$

1,2,3

$3\times10^0 \rightarrow 3$

$+\ 2\times10^1 \rightarrow 20$

$1\times10^2 \rightarrow 100$

123

$(53)_{10} = (\ )_3$ → (0, 1, 2)

ans = 2 + 20 + 200 + 1000

n → 53

3	53	2×10^0
3	17	2×10^1
3	5	2×10^2
3	1	1×10^3
	0	

2
20
200
1000
1222

n = ~~53~~ ~~17~~ ~~5~~ ~~1~~ 0

b = 3

pow = 1 ~~10~~ ~~1000~~ 10000

while (n > 0)

digit = n % b

n = n / b

ans = ans + (digit × pow)

pow = pow × 10

ans = 1222

Any Base System to Decimal :

Decimal Number Expansion (Positional Value Concept)

Decimal (Base 10)

$123 \rightarrow 1\times 10^2 + 2\times 10^1 + 3\times 10^0$

Base

$(1101101)_2 = (\ \ )_{10}$

2

$1101101 = 1\times 2^0 + \cancel{0\times 2^1} + 1\times 2^2 + \cancel{0\times 2^3} + 1\times 2^4 + 1\times 2^5$

$= 1 + 4 + 16 + 32$

$= 21 + 32 = 53$

$(1222)_3 = (\quad)_{10}$

$2\times3^0 + 2\times3^1 + 2\times3^2$

$n \to 1222$

$b \to 3$

ans

pow = 1

$n \to \cancel{1222}\ \cancel{122}\ \cancel{12}\ \cancel{1}$ $\quad 1/10 \to 0$

$b \to 3$

$pow \to \cancel{1}\ \cancel{3}\ \cancel{3^2}\ \cancel{3^3}\ 3^4$

$ans = \cancel{2+6+18+27}$

digit = ~~2~~ 1

ans → 53

```
while (n>0) {
    digit = n%10
    n = n/10
    ans = ans + (digit x pow)
    pow = pow x b
}
```